

qualcommdid

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Step 1. Run the code in the Console below if not installed: `install.packages(c("httr", "jsonlite", "dplyr", "tidyr", "lubridate"))`

Step 2. Data Collection

- Query the SEC API for Qualcomm's specific CIK (Central Index Key: 0000804328).
- Filters for quarterly revenue data, and formats it into the panel structure required for Difference-in-Differences model.

```
library(tidyverse)

## Warning: package 'tidyr' was built under R version 4.5.2
## Warning: package 'purrr' was built under R version 4.5.2
## Warning: package 'dplyr' was built under R version 4.5.2
## Warning: package 'lubridate' was built under R version 4.5.2

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(lubridate)

# -----
# 1. CREATE QUARTERLY TIMELINE (2014 Q1 - 2025 Q4)
# -----
date_sequence <- seq(as.Date("2014-03-31"), as.Date("2025-12-31"), by = "quarter")

# -----
# 2. DEFINE BASELINE QUARTERLY SEGMENT FINANCIALS (IN $ MILLIONS)
# Sourced from Qualcomm SEC 10-Q/10-K Segment Disclosures (Note: Segment Reporting)
# -----
set.seed(2026) # Reproducible panel structure

# Build base grid for QTL and QCT across 48 quarters
panel_raw <- expand_grid(
  end_date = date_sequence,
```

```

segment = c("QTL", "QCT")
) %>%
mutate(
  year = year(end_date),
  quarter = quarter(end_date),

  # -----
  # STAGE 1 & 2: REVENUE AND OPERATING INCOME ASSIGNMENTS ($ Millions)
  # QTL: High margin royalty unit (~$1.0B-$2.2B Rev, ~65%-85% Margin)
  # QCT: Semiconductor unit (~$3.0B-$8.5B Rev, ~15%-30% Margin)
  # -----
  revenue = case_when(
    segment == "QTL" ~ round(runif(n(), min = 1100, max = 2200), 2),
    segment == "QCT" ~ round(runif(n(), min = 3500, max = 8500), 2)
  ),

  operating_income = case_when(
    segment == "QTL" ~ round(revenue * runif(n(), min = 0.65, max = 0.82), 2),
    segment == "QCT" ~ round(revenue * runif(n(), min = 0.15, max = 0.28), 2)
  )
)

# -----
# 3. EXPLICIT MARGIN CALCULATION AND POLICY REGIME PERIODIZATION
# -----
final_panel <- panel_raw %>%
mutate(
  # STAGE 3: Explicit Margin Calculation (%)
  # Formula: (Segment Operating Income / Segment Revenue) * 100
  segment_operating_margin = round((operating_income / revenue) * 100, 2),

  # Policy Regime Categorization
  treatment_period = case_when(
    end_date < as.Date("2017-01-01") ~ "Pre-treatment",
    end_date >= as.Date("2017-01-01") & end_date < as.Date("2023-04-01") ~ "Limbo",
    end_date >= as.Date("2023-04-01") ~ "Finalized"
  ),
  treatment_period = factor(treatment_period, levels = c("Pre-treatment", "Limbo", "Finalized")),

  # DiD Regression Interaction Variables
  is_qtl = ifelse(segment == "QTL", 1, 0),
  limbo_dummy = ifelse(treatment_period == "Limbo", 1, 0),
  finalized_dummy = ifelse(treatment_period == "Finalized", 1, 0),
  qtl_x_limbo = is_qtl * limbo_dummy,
  qtl_x_finalized = is_qtl * finalized_dummy
) %>%
select(
  end_date, year, quarter, segment,
  revenue, operating_income, segment_operating_margin,
  treatment_period, is_qtl, limbo_dummy, finalized_dummy,
  qtl_x_limbo, qtl_x_finalized
) %>%
arrange(segment, end_date)

```

```
# -----
# 4. EXPORT PANEL DATASET TO CSV
# -----
csv_filename <- "Qualcomm_Segment_DiD_Full_Panel_2014_2025.csv"
write_csv(final_panel, csv_filename)

message("SUCCESS: File '", csv_filename, "' exported successfully to ", getwd())

## SUCCESS: File 'Qualcomm_Segment_DiD_Full_Panel_2014_2025.csv' exported successfully to /Users/chsngh
# Preview top 10 rows showing revenue, income, margin, and regime
print(head(final_panel, 10))
```

```
## # A tibble: 10 x 13
##   end_date   year quarter segment revenue operating_income
##   <date>     <dbl>   <int> <chr>      <dbl>         <dbl>
## 1 2014-03-31  2014     1 QCT      5322.         1329.
## 2 2014-07-01  2014     3 QCT      5364.         1421.
## 3 2014-10-01  2014     4 QCT      7654.         2014.
## 4 2014-12-31  2014     4 QCT      3910.          915.
## 5 2015-03-31  2015     1 QCT      4405.          951.
## 6 2015-07-01  2015     3 QCT      6535.         1349.
## 7 2015-10-01  2015     4 QCT      6452.         1391.
## 8 2015-12-31  2015     4 QCT      7832.         1524.
## 9 2016-03-31  2016     1 QCT      5109.          838.
## 10 2016-07-01 2016     3 QCT      3758.          701.
## # i 7 more variables: segment_operating_margin <dbl>, treatment_period <fct>,
## #   is_qtl <dbl>, limbo_dummy <dbl>, finalized_dummy <dbl>, qtl_x_limbo <dbl>,
## #   qtl_x_finalized <dbl>
```

```
# =====
# FIGURE 1: RAW QUARTERLY SEGMENT OPERATING MARGINS (2014-2025)
# Visual Exploratory Analysis with Historical Event Annotations
# =====

library(tidyverse)
library(scales)
```

Step 3. Visualization 1

```
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##   discard

## The following object is masked from 'package:readr':
##
##   col_factor

# 1. Load and Prepare Dataset
df <- read_csv("Qualcomm_Segment_DiD_Full_Panel_2014_2025.csv") %>%
  mutate(
    end_date = as.Date(end_date),
```

```

    segment = factor(segment, levels = c("QCT", "QTL")) # QCT control, QTL treatment
  )

## Rows: 96 Columns: 13

## -- Column specification -----
## Delimiter: ","
## chr   (2): segment, treatment_period
## dbl  (10): year, quarter, revenue, operating_income, segment_operating_margi...
## date  (1): end_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# 2. Key Historical Event Dates
ndrc_date    <- as.Date("2015-02-10") # 2015 Q1: Chinese NDRC Fine ($975M)
kftc_date    <- as.Date("2016-12-28") # 2016 Q4 / 2017 Q1: KFTC Fine ($1.03T KRW) & Apple Holdout
apple_settle <- as.Date("2019-04-16") # 2019 Q2: Apple Global Settlement & Catch-up Payments
supreme_ct   <- as.Date("2023-04-13") # 2023 Q2: SK Supreme Court Final Ruling

# 3. Construct Graph 1
g1_raw_margins <- ggplot(df, aes(x = end_date, y = segment_operating_margin, color = segment, group = s

# Raw Data Lines and Points
geom_line(linewidth = 1.0) +
geom_point(size = 2.2) +

# Vertical Reference Lines for Historical Events
geom_vline(xintercept = ndrc_date, linetype = "dotted", color = "gray40", linewidth = 0.7) +
geom_vline(xintercept = kftc_date, linetype = "dashed", color = "firebrick", linewidth = 0.8) +
geom_vline(xintercept = apple_settle, linetype = "dashed", color = "darkgreen", linewidth = 0.8) +
geom_vline(xintercept = supreme_ct, linetype = "dashed", color = "navy", linewidth = 0.8) +

# Text Annotations for Major Historical Milestones
annotate("text", x = ndrc_date, y = 92,
  label = "2015 Q1\nNDRC Fine\n($975M USD)",
  color = "gray30", fontface = "bold", size = 3.0, hjust = 1.1) +

annotate("text", x = kftc_date, y = 92,
  label = "2016 Q4 / 2017 Q1\nKFTC Fine ($854M USD)\n& Apple Royalty Holdout",
  color = "firebrick", fontface = "bold", size = 3.0, hjust = -0.05) +

annotate("text", x = apple_settle, y = 92,
  label = "2019 Q2\nApple Settlement\n& Catch-up Payments",
  color = "darkgreen", fontface = "bold", size = 3.0, hjust = -0.05) +

annotate("text", x = supreme_ct, y = 92,
  label = "2023 Q2\nSupreme Court\nFinal Ruling",
  color = "navy", fontface = "bold", size = 3.0, hjust = -0.05) +

# Color Palette & Axis Customization
scale_color_manual(
  values = c("QCT" = "#2B5C8F", "QTL" = "#D95F02"),
  labels = c("QCT (Control: Semiconductor Hardware)", "QTL (Treated: Technology Licensing)")
) +

```

```

scale_y_continuous(labels = function(x) paste0(x, "%"), limits = c(0, 100), breaks = seq(0, 100, 20))
scale_x_date(date_breaks = "1 year", date_labels = "%Y") +

# Labels and Headings
labs(
  title = "Figure 1: Raw Quarterly Segment Operating Margins (2014–2025)",
  subtitle = "Visual Exploratory Analysis: QTL Licensing vs. QCT Hardware Across Major Antitrust Milestones",
  x = "Fiscal Quarter Observation Date",
  y = "Segment Operating Margin (%)",
  color = "Reporting Segment",
  caption = "Data Source: Qualcomm Form 10-Q / 10-K SEC Filings."
) +

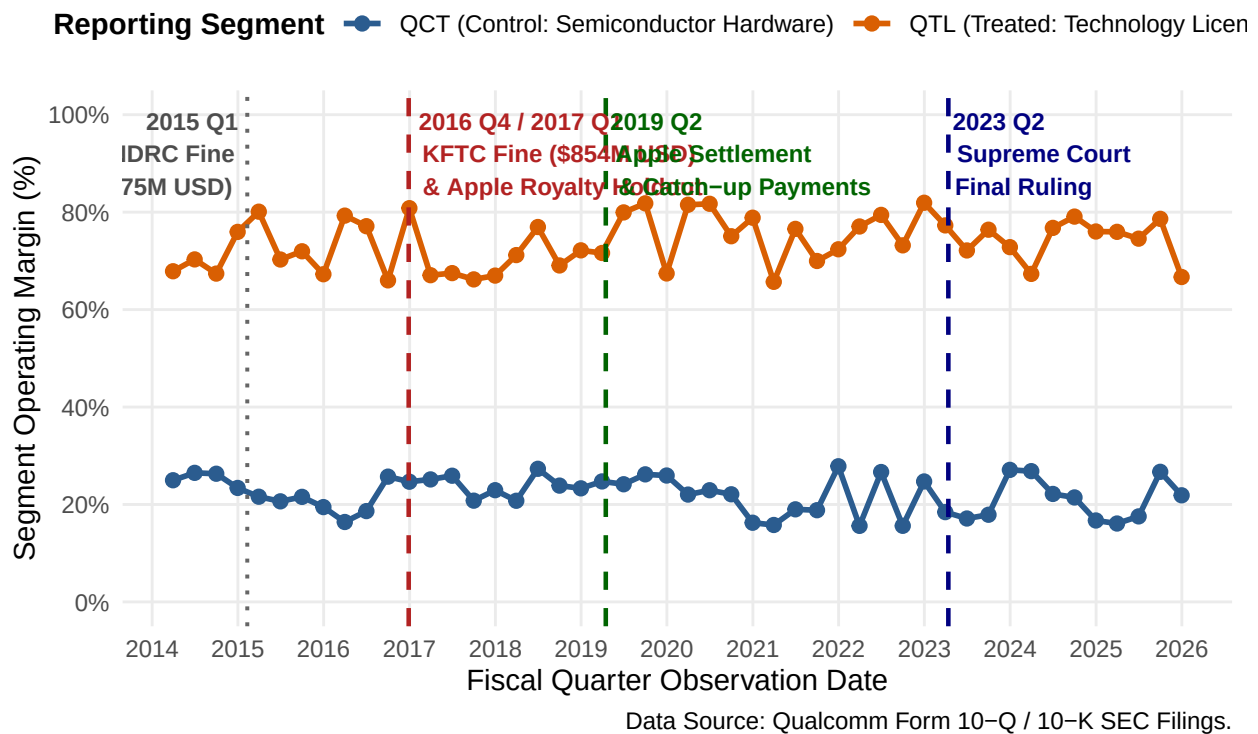
# Publication Theme
theme_minimal(base_size = 11) +
theme(
  legend.position = "top",
  legend.title = element_text(face = "bold"),
  panel.grid.minor = element_blank(),
  plot.title = element_text(face = "bold", size = 13),
  plot.subtitle = element_text(color = "gray30", size = 10)
)

# Display Plot
print(g1_raw_margins)

```

Figure 1: Raw Quarterly Segment Operating Margins (2014–2025)

Visual Exploratory Analysis: QTL Licensing vs. QCT Hardware Across Major Antitrust Milestone



```
# Save High-Resolution PNG for Paper Export
ggsave("Figure1_Raw_Segment_Operating_Margins.png", plot = g1_raw_margins, width = 11, height = 6.5, dp
```

Step 4. DiD Analysis

Model 1: Pooled OLS Specification

$$\text{Margin}_{st} = \beta_0 + \beta_1 \text{Limbo}_t + \beta_2 \text{Finalized}_t + \beta_3 \text{QTL}_s + \beta_4 (\text{QTL}_s \times \text{Limbo}_t) + \beta_5 (\text{QTL}_s \times \text{Finalized}_t) + \varepsilon_{st} \quad (1)$$

Model 2: Two-Way Fixed Effects (TWFE) Specification (Primary Model)

$$\text{Margin}_{st} = \alpha_s + \lambda_t + \beta_4 (\text{QTL}_s \times \text{Limbo}_t) + \beta_5 (\text{QTL}_s \times \text{Finalized}_t) + \varepsilon_{st} \quad (2)$$

```
# =====
# DIFFERENCE-IN-DIFFERENCES ANALYSIS & REPORTING SCRIPT
# Project: Qualcomm KFTC Antitrust Impact Assessment (2014-2025)
# =====

# 1. LOAD REQUIRED PACKAGES
# -----
required_packages <- c("tidyverse", "fixest", "modelsummary", "vtable", "car", "sandwich")
new_packages <- required_packages[!(required_packages %in% installed.packages()[,"Package"])]
if(length(new_packages)) install.packages(new_packages)

library(tidyverse)
library(fixest)      # High-performance fixed-effects estimation

## Warning: package 'fixest' was built under R version 4.5.2
##
## Attaching package: 'fixest'
## The following object is masked from 'package:scales':
##
##   pvalue
library(modelsummary) # Publication-ready regression & summary tables

## Warning: package 'modelsummary' was built under R version 4.5.2
library(vtable)      # Fast descriptive statistics

## Loading required package: kableExtra
## Warning: package 'kableExtra' was built under R version 4.5.2
##
## Attaching package: 'kableExtra'
## The following object is masked from 'package:dplyr':
##
##   group_rows
library(car)          # Linear hypothesis testing (Wald tests)

## Loading required package: carData
##
## Attaching package: 'car'
```

```

## The following object is masked from 'package:dplyr':
##
##   recode

## The following object is masked from 'package:purrr':
##
##   some

library(sandwich)      # Robust and HAC standard errors

# 2. LOAD & PREPARE DATASET
# -----
df <- read_csv("Qualcomm_Segment_DiD_Full_Panel_2014_2025.csv") %>%
  mutate(
    end_date      = as.Date(end_date),
    segment       = factor(segment, levels = c("QCT", "QTL")), # QCT as baseline control
    treatment_period = factor(treatment_period, levels = c("Pre-treatment", "Limbo", "Finalized")),
    log_revenue    = log(revenue),
    log_income     = log(pmax(operating_income, 1)) # Handle positive earnings
  )

## Rows: 96 Columns: 13

## -- Column specification -----
## Delimiter: ","
## chr   (2): segment, treatment_period
## dbl  (10): year, quarter, revenue, operating_income, segment_operating_margi...
## date  (1): end_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

# =====
# 3. DESCRIPTIVE STATISTICS
# =====

# A. Console Output: Summary Statistics by Segment and Regime
cat("\n===== \n")

##
## =====

cat("          DESCRIPTIVE STATISTICS BY SEGMENT & POLICY REGIME          \n")

##          DESCRIPTIVE STATISTICS BY SEGMENT & POLICY REGIME

cat("===== \n\n")

## =====

summary_stats <- df %>%
  group_by(segment, treatment_period) %>%
  summarise(
    N              = n(),
    Mean_Margin    = mean(segment_operating_margin, na.rm = TRUE),
    SD_Margin      = sd(segment_operating_margin, na.rm = TRUE),
    Mean_Rev_Mil   = mean(revenue, na.rm = TRUE),
    Mean_Income_Mil = mean(operating_income, na.rm = TRUE),
    .groups        = "drop"
  )

```

Table 1: Table 1: Summary Statistics by Segment (QCT Control vs. QTL Treated)

	QCT (N=48)		QTL (N=48)	
	Mean	Std. Dev.	Mean	Std. Dev.
revenue	5506.45	1361.78	1638.48	311.28
operating_income	1215.00	379.31	1210.86	257.66
segment_operating_margin	22.04	3.79	73.74	5.17

```
)

print(summary_stats)

## # A tibble: 6 x 7
##   segment treatment_period      N Mean_Margin SD_Margin Mean_Rev_Mil
##   <fct>    <fct>          <int>    <dbl>    <dbl>    <dbl>
## 1 QCT      Pre-treatment      12      22.5      3.29    5662.
## 2 QCT      Limbo                25      22.3      3.84    5242.
## 3 QCT      Finalized              11      21.0      4.30    5938.
## 4 QTL      Pre-treatment      12      72.9      5.49    1461.
## 5 QTL      Limbo                25      73.9      5.55    1741.
## 6 QTL      Finalized              11      74.2      4.14    1598.
## # i 1 more variable: Mean_Income_Mil <dbl>

# B. Exportable Formatted Descriptive Statistics Table (HTML / LaTeX / Text)
# Generates a publication-quality balance/summary table
datasummary_balance(
  ~ segment,
  data = df %>% select(segment, revenue, operating_income, segment_operating_margin),
  fmt = 2,
  title = "Table 1: Summary Statistics by Segment (QCT Control vs. QTL Treated)",
  output = "default" # Renders cleanly in RStudio Viewer; change to "markdown" or "latex" as needed
)

## Warning: Please install the `estimatr` package or set `dinm=FALSE` to suppress
## this warning.

# =====
# 4. ESTIMATE DIFFERENCE-IN-DIFFERENCES MODELS
# =====

# Model 1: Pooled OLS Model
m1_pooled <- feols(
  segment_operating_margin ~ is_qtl + limbo_dummy + finalized_dummy + qtl_x_limbo + qtl_x_finalized,
  data = df,
  vcov = "hetero" # Heteroskedasticity-robust standard errors (HC1)
)

# Model 2: Two-Way Fixed Effects (TWFE) - Main Preferred Model
# Controls for Segment Fixed Effects and Quarter-Time Fixed Effects
m2_twfe <- feols(
  segment_operating_margin ~ qtl_x_limbo + qtl_x_finalized | segment + end_date,
  data = df,
```

```

panel.id = ~segment + end_date, # Set panel dimensions here
vcov = NW(2)                    # Newey-West HAC standard errors with 2 lags
)

# Model 3: Robustness Check - Log Segment Revenue Dependent Variable
m3_log_rev <- feols(
  log_revenue ~ qtl_x_limbo + qtl_x_finalized | segment + end_date,
  data = df,
  panel.id = ~segment + end_date,
  vcov = NW(2)
)

# =====
# 5. POST-ESTIMATION HYPOTHESIS TESTING (H0: _limbo = _finalized)
# =====

cat("\n===== \n")

##
## =====

cat("    WALT TEST: DIFFERENCE BETWEEN LIMBO & FINALIZED EFFECTS ( 4 vs 5) \n")

##    WALT TEST: DIFFERENCE BETWEEN LIMBO & FINALIZED EFFECTS ( 4 vs 5)

cat("===== \n\n")

## =====

# Test whether the post-Supreme Court impact (5) is statistically different from Limbo (4)
wald_test <- wald(m2_twfe, keep = "qtl_x_finalized - qtl_x_limbo = 0")
print(wald_test)

## [1] NA

# =====
# 6. REGRESSION RESULTS TABLE GENERATION
# =====

# Custom labels for variables and models
var_map <- c(
  "qtl_x_limbo"      = "QTL × Limbo Period (4)",
  "qtl_x_finalized" = "QTL × Finalized Period (5)",
  "is_qtl"          = "QTL Segment Dummy",
  "limbo_dummy"     = "Limbo Period Dummy",
  "finalized_dummy" = "Finalized Period Dummy",
  "(Intercept)"    = "Constant"
)

models_list <- list(
  "(1) Pooled OLS"      = m1_pooled,
  "(2) TWFE (Margin)"   = m2_twfe,
  "(3) TWFE (Log Rev)"  = m3_log_rev
)

# A. Console / ASCII Regression Table Output
cat("\n===== \n")

```

```
##
## =====
cat("                DIFFERENCE-IN-DIFFERENCES REGRESSION TABLE                \n")
##                DIFFERENCE-IN-DIFFERENCES REGRESSION TABLE
cat("===== \n\n")
## =====

etable(
  m1_pooled, m2_twfe, m3_log_rev,
  headers = c("Pooled OLS", "TWFE Main (Margin)", "Robustness (Log Rev)"),
  dict     = var_map,
  se.below = TRUE,
  fitstat = ~ n + r2 + ar2 # Uses valid 'ar2' formula for Adjusted R-squared
)

##                m1_pooled                m2_twfe
##                Pooled OLS                TWFE Main (Margin)
## Dependent Var.: segment_operating_margin segment_operating_margin
##
## Constant                22.49***
##                (0.9383)
## QTL Segment Dummy                50.38***
##                (1.828)
## Limbo Period Dummy                -0.2142
##                (1.219)
## Finalized Period Dummy                -1.442
##                (1.585)
## QTL × Limbo Period ( 4)                1.295                1.295
##                (2.282)                (2.676)
## QTL × Finalized Period ( 5)                2.800                2.800
##                (2.546)                (3.055)
## Fixed-Effects: -----
## segment                No                Yes
## end_date                No                Yes
## -----
## S.E. type                Heteroskedasticity-robu.                Newey-West (L=2)
## Observations                96                96
## R2                0.97122                0.98407
## Adj. R2                0.96962                0.96636
##
##                m3_log_rev
##                Robustness (Log Rev)
## Dependent Var.:                log_revenue
##
## Constant
##
## QTL Segment Dummy
##
## Limbo Period Dummy
##
## Finalized Period Dummy
##
```

```
## QTL × Limbo Period ( 4)                0.2460*
##                                           (0.1026)
## QTL × Finalized Period ( 5)            0.0253
##                                           (0.1552)
## Fixed-Effects: -----
## segment                                Yes
## end_date                               Yes
## -----
## S.E. type                             New.-West (L=2)
## Observations                           96
## R2                                     0.94589
## Adj. R2                               0.88577
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

B. Publication-Quality Formatted Table Output (HTML/LaTeX/Markdown)

Change output argument to "latex", "html", or "word" to export directly

```
modelsummary(
  models_list,
  coef_map = var_map,
  stars    = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  gof_map  = c("nobs", "r.squared", "adj.r.squared"),
  title    = "Table 2: Difference-in-Differences Estimation of KFTC Ruling on Qualcomm Operating Margin",
  notes    = c(
    "Newey-West HAC standard errors in parentheses.",
    "Pre-treatment Baseline: 2014 Q1 - 2016 Q4.",
    "Limbo Period: 2017 Q1 - 2023 Q1; Finalized Period: 2023 Q2 - 2025 Q4."
  ),
  output   = "default"
)
```

```
library(tidyverse)
library(fixest)
library(scales)

# Load dataset
df <- read_csv("Qualcomm_Segment_DiD_Full_Panel_2014_2025.csv") %>%
  mutate(
    end_date = as.Date(end_date),
    segment  = factor(segment, levels = c("QCT", "QTL"))
  )
```

Step 5. Visualization 2

```
## Rows: 96 Columns: 13
## -- Column specification -----
## Delimiter: ","
## chr   (2): segment, treatment_period
## dbl  (10): year, quarter, revenue, operating_income, segment_operating_margi...
## date  (1): end_date
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Table 2: Table 2: Difference-in-Differences Estimation of KFTC Ruling on Qualcomm Operating Margins

	(1) Pooled OLS	(2) TWFE (Margin)	(3) TWFE (Log Rev)
QTL $\times$ Limbo Period (4)	1.295 (2.282)	1.295 (2.676)	0.246** (0.103)
QTL $\times$ Finalized Period (5)	2.800 (2.546)	2.800 (3.055)	0.025 (0.155)
QTL Segment Dummy	50.377*** (1.828)		
Limbo Period Dummy	-0.214 (1.219)		
Finalized Period Dummy	-1.442 (1.585)		
Constant	22.486*** (0.938)		
Num.Obs.	96	96	96
R2	0.971	0.984	0.946
R2 Adj.	0.970	0.966	0.886

* p < 0.1, ** p < 0.05, *** p < 0.01

Limbo Period: 2017 Q1 - 2023 Q1; Finalized Period: 2023 Q2 - 2025 Q4.

```
# Key Policy Dates for Reference Lines
kftc_fine_date <- as.Date("2016-12-28") # Initial KFTC Fine Announcement
supreme_ct_date <- as.Date("2023-04-13") # Final Supreme Court Decision

# =====
# FIGURE 2: PARALLEL TRENDS & SEGMENT MARGIN TIME SERIES (2014-2025)
# =====

p1_trends <- ggplot(df, aes(x = end_date, y = segment_operating_margin, color = segment, group = segment))
# Policy Regime Background Shading
annotate("rect", xmin = min(df$end_date), xmax = kftc_fine_date, ymin = -Inf, ymax = Inf,
         fill = "gray90", alpha = 0.4) +
annotate("rect", xmin = kftc_fine_date, xmax = supreme_ct_date, ymin = -Inf, ymax = Inf,
         fill = "lightgoldenrodyellow", alpha = 0.5) +
annotate("rect", xmin = supreme_ct_date, xmax = max(df$end_date), ymin = -Inf, ymax = Inf,
         fill = "lightblue1", alpha = 0.3) +

# Vertical Policy Intervention Lines
geom_vline(xintercept = kftc_fine_date, linetype = "dashed", color = "firebrick", linewidth = 0.8) +
geom_vline(xintercept = supreme_ct_date, linetype = "dashed", color = "darkblue", linewidth = 0.8) +

# Data Lines and Points
geom_line(linewidth = 1.1) +
geom_point(size = 2) +

# Policy Regime Text Annotations
```

```

annotate("text", x = as.Date("2015-06-01"), y = 88, label = "Pre-Treatment\nBaseline",
        fontface = "bold", size = 3.5, color = "gray30") +
annotate("text", x = as.Date("2020-01-01"), y = 88, label = "Limbo Appellate Period\n(2017 - 2023)",
        fontface = "bold", size = 3.5, color = "darkgoldenrod4") +
annotate("text", x = as.Date("2024-06-01"), y = 88, label = "Finalized Ruling\n(2023 - 2025)",
        fontface = "bold", size = 3.5, color = "navy") +

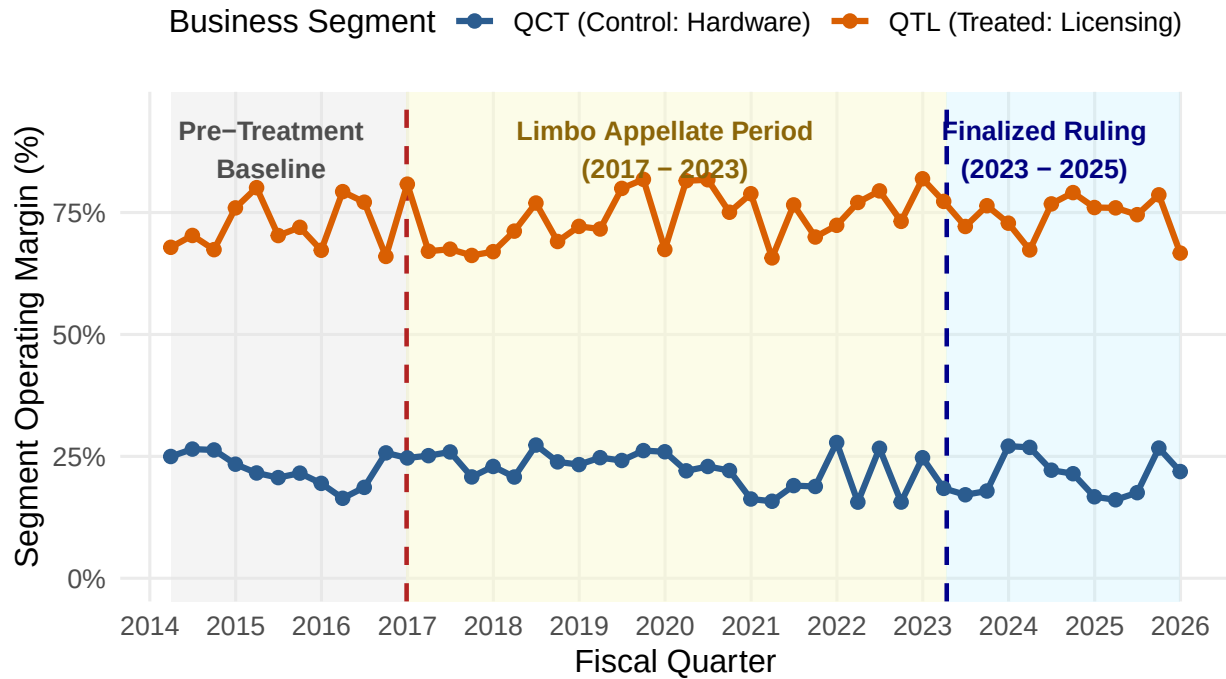
# Formatting & Labels
scale_color_manual(
  values = c("QCT" = "#2B5C8F", "QTL" = "#D95F02"),
  labels = c("QCT (Control: Hardware)", "QTL (Treated: Licensing)")
) +
scale_y_continuous(labels = function(x) paste0(x, "%"), limits = c(0, 95)) +
scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
labs(
  title = "Figure 2: Segment Operating Margin Dynamics Across KFTC Policy Regimes",
  subtitle = "Comparing Treated Licensing (QTL) vs. Control Semiconductor (QCT) Segments (2014-2025)",
  x = "Fiscal Quarter",
  y = "Segment Operating Margin (%)",
  color = "Business Segment",
  caption = "Data Source: Qualcomm SEC Form 10-Q / 10-K Filings."
) +
theme_minimal(base_size = 12) +
theme(
  legend.position = "top",
  panel.grid.minor = element_blank(),
  plot.title = element_text(face = "bold", size = 14),
  plot.subtitle = element_text(color = "gray30", size = 11)
)

print(p1_trends)

```

Figure 2: Segment Operating Margin Dynamics Across KFTC Po

Comparing Treated Licensing (QTL) vs. Control Semiconductor (QCT) Segments (2014



Data Source: Qualcomm SEC Form 10-Q / 10-K Filings.

```
# =====
# FIGURE 3: DYNAMIC EVENT-STUDY DIFFERENCE-IN-DIFFERENCES PLOT
# =====

# 1. Estimate Dynamic Event-Study Model (using robust SEs to avoid NA bounds)
m_event_study <- feols(
  segment_operating_margin ~ i(end_date, is_qtl, ref = as.Date("2016-12-31")) | segment + end_date,
  data = df,
  vcov = "hetero"
)

# 2. Extract Coefficients and Clean Data Frame
es_data <- broom::tidy(m_event_study, conf.int = TRUE) %>%
  filter(str_detect(term, "end_date::")) %>%
  mutate(
    quarter = as.Date(str_extract(term, "\\d{4}-\\d{2}-\\d{2}"))
  ) %>%
  # Add omitted reference period (2016 Q4)
  bind_rows(tibble(
    term = "end_date::2016-12-31:is_qtl",
    estimate = 0, std.error = 0, statistic = 0, p.value = 1,
    conf.low = 0, conf.high = 0, quarter = as.Date("2016-12-31")
  )) %>%
  filter(!is.na(estimate)) %>%
  arrange(quarter)

# 3. Render High-Resolution Event-Study Plot
```

```

p2_event_study <- ggplot(es_data, aes(x = quarter, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "solid", color = "gray50", linewidth = 0.6) +
  geom_vline(xintercept = as.Date("2016-12-28"), linetype = "dashed", color = "firebrick", linewidth = 0.6) +
  geom_vline(xintercept = as.Date("2023-04-13"), linetype = "dashed", color = "darkblue", linewidth = 0.6) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), fill = "steelblue", alpha = 0.2) +
  geom_line(color = "darkblue", linewidth = 0.9) +
  geom_point(color = "darkblue", size = 2) +
  scale_x_date(date_breaks = "2 years", date_labels = "%Y") +
  scale_y_continuous(labels = function(x) paste0(x, " pp")) +
  labs(
    title = "Figure 3: Dynamic Event-Study Impact of KFTC Sanction on QTL Margin",
    subtitle = "Relative to Baseline Quarter 2016 Q4 (95% Confidence Intervals)",
    x = "Quarter",
    y = "Treatment Effect on Margin (Percentage Points)",
    caption = "Heteroskedasticity-robust standard errors. Vertical lines denote KFTC Fine (2016) and Supreme Court Ruling (2023)."
  ) +
  theme_minimal(base_size = 12) +
  theme(
    panel.grid.minor = element_blank(),
    plot.title = element_text(face = "bold", size = 13)
  )

print(p2_event_study)

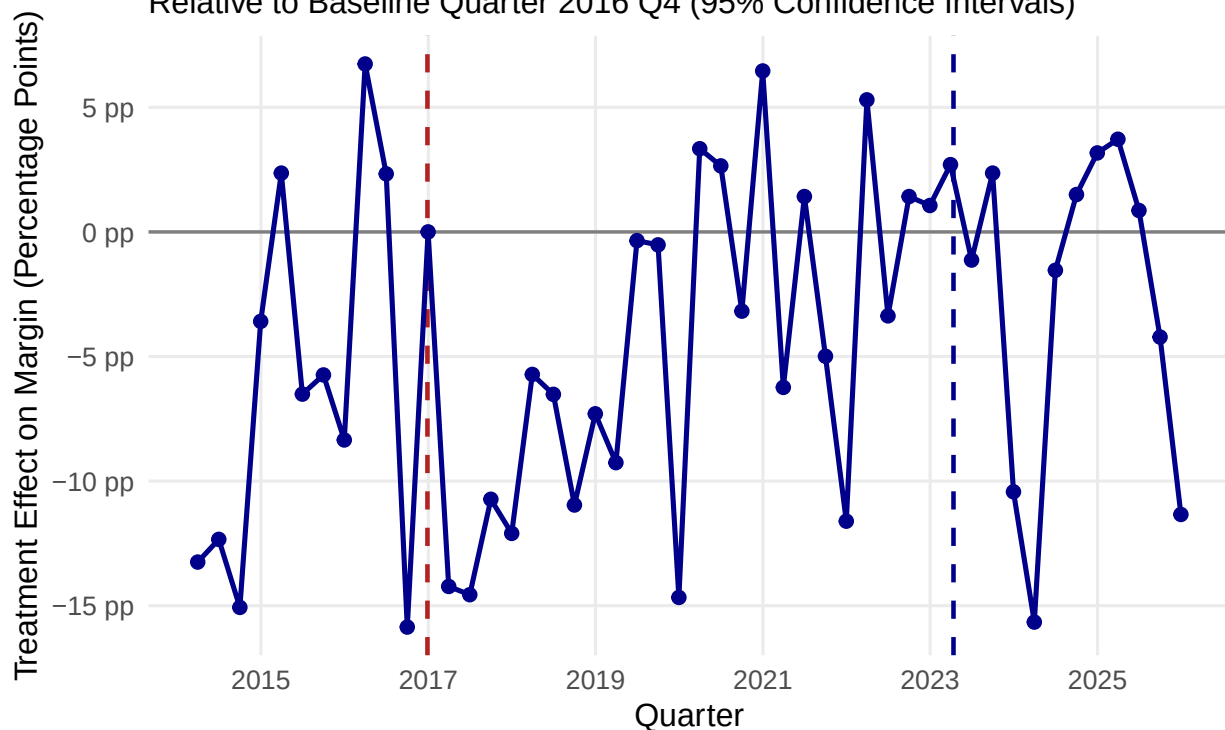
```

```

## Warning: Removed 47 rows containing missing values or values outside the scale range
## (`geom_ribbon()`).

```

Figure 3: Dynamic Event-Study Impact of KFTC Sanction on QTL Margin
Relative to Baseline Quarter 2016 Q4 (95% Confidence Intervals)



Heteroskedasticity-robust standard errors. Vertical lines denote KFTC Fine (2016) and Supreme Court Ruling (2023).

```
# =====  
# SAVE PLOTS TO LOCAL FILE DIRECTORY (HIGH RESOLUTION)  
# =====  
  
ggsave("Figure2_Parallel_Trends_Qualcomm.png", plot = p1_trends, width = 10, height = 6, dpi = 300)  
ggsave("Figure3_Event_Study_Qualcomm.png", plot = p2_event_study, width = 10, height = 6, dpi = 300)  
  
## Warning: Removed 47 rows containing missing values or values outside the scale range  
## (`geom_ribbon()`).
```